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WiFiNINA Library Examples

The WiFi library for boards with a NINA WiFi module. Works with the MKR WiFi 1010, MKR Vidor 4000, Uno WiFi Rev2, Nano 33 IoT and the Nano RP2040 Connect.

Last revision • 09/30/2025

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The [WiFiNINA library](#) is designed for Arduino boards using a [NINA W-10 series module](#). In this article you will find a series of examples that can be uploaded to your board.

You can also visit the [WiFiNINA GitHub repository](#) to learn more about this library.

Hardware Required

[Arduino MKR WiFi 1010](#)

[Arduino MKR VIDOR 4000](#)

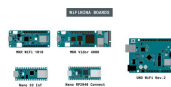
[Arduino UNO WiFi Rev.2](#)

[Arduino Nano 33 IoT](#)

[Arduino Nano RP2040 Connect](#)

Circuit

Most examples in this article uses no external circuit, only the board itself is required.



WiFiNINA
compatible
boards.

Please note: these three boards use dedicated pins to communicate and select the WiFi

 module, therefore you have no restriction in the usage of the available digital pins connected to the header pins.

Examples

WiFiNINA AP Simple Web Server

In this example, a simple web server lets you blink an LED via the web. This example uses the `beginAP()` function to set up an access point without relying on a local WiFi network. This example will print the IP address of your WiFi module to the Arduino Software (IDE) serial monitor. Once you know the IP address of our board, you can open that address in a web browser to turn on and off the LED on pin 9.

If the IP address of your shield is `yourAddress`:

<http://yourAddress/H>
turns the LED on

<http://yourAddress/L>
turns it off

The default address of the board in AP mode is 192.168.4.1. When you load this sketch, the WiFi module creates an Access Point with the name specified as SSID in `arduino_secrets.h`. Connect to it using the password specified as PASS.

```
1  /*
2
3  WiFi Web Serve
4
5  A simple web s
6
7  This sketch wi
8
9  It will then l
10
11  to the Serial
12
13  to turn on and
14
15  If the IP addr
16
17  http://yourA
18
19  http://yourA
20
21  created 25 Nov
22
23  by Tom Igoe
24
25  adapted to WiF
26
27  */
28
```

WiFiNINA Connect No Encryption

This example shows you how to connect to an open (not

network with one of the boards that support this library. Your Arduino Software (IDE) serial monitor will provide information about the connection once it has connected.

You should have access to a 802.11b/g wireless network that connects to the internet for this example. You will need to change the network settings in the sketch to correspond to your particular networks SSID.

```
1  /*
2
3   This example co
4
5   Then it prints
6
7   the IP address
8
9   created 13 July
10
11  by dlf (Metodo2
12
13  modified 31 May
14
15  by Tom Igoe
16
17  */
18  #include <SPI.h>
19  #include <WiFI
20
21  #include "arduin
22  //please en
23  char ssid[] = SE
24  int status = WL_
25
26  void setup() {
27
28    //Initialize s
```

WiFiNINA Connect With WEP

This example shows you how to connect to a WEP encrypted 802.11b/g network with one of the boards that support this library. Your Arduino Software (IDE) serial monitor will provide information about the connection once it has connected.

You should have access to a 802.11b/g wireless network that connects to the internet for this example. You will need to change the network settings in the sketch to correspond to your particular networks SSID.

WEP network passwords are hexadecimal strings known as keys. A WEP network can have 4 different keys; each key is assigned a "Key Index" value. For WEP encrypted networks, you need the SSID, the key, and key number.

```
1  /*
2
3  This example co
4
5  Then it prints
6
7  the IP address
8
9  If you use 40-b
10
11 and the charact
12
13 e.g. for 40-bi
14
15 (too short) and
16
17 hexadecimal cha
18
19 For 128-bit, yo
20
21 D0D0DEADF00DABB
22
23 all in the 0-9,
24
25 created 13 July
26
27 by dlf (Metodo2
28
```

WifiNINA Connect With WPA

This example shows you how to connect to a WPA2 Personal encrypted 802.11b/g network with one of the boards that support this library. Your Arduino Software (IDE) serial monitor will provide information about the connection once it has connected.

You should have access to a 802.11b/g wireless network

need to change the network settings in the sketch to correspond to your particular networks SSID.

For networks using WPA/WPA2 Personal encryption, you need the SSID and password. The module will not connect to networks using WPA2 Enterprise encryption.

```
1  /*
2
3   This example co
4
5   Then it prints
6
7   the IP address
8
9   created 13 July
10
11  by dlf (Metodo2
12
13  modified 31 May
14
15  by Tom Igoe
16
17  */
18  #include <SPI.h>
19  #include <Wi-FiNINA.h>
20
21  #include "arduino_wifi_nina.h"
22  //please en
23  char ssid[] = SE
24  char pass[] = SE
25  int status = WL_
26
27  void setup() {
28
```

WifiNINA Scan Networks

This example scans for 802.11b/g network with one of the boards that support this library. Your Arduino Software (IDE) serial monitor will print out information about the board and the networks it can see. It will not connect to a network.

Open your Arduino Software (IDE) serial monitor to view the networks the WiFi module can see. The shield may not see as many networks as your computer.

```
1  /*
2
3   This example p
4
5   scans for avail
6
7   Every ten secon
8
9   connect to any
10
11  Circuit:
12
13   * Board with NI
14
15   created 13 July
16
17   by dlf (Metodo2
18
19   modified 21 Jun
20
21   by Tom Igoe and
22
23   */
24
25  #include <SPI.h>
26  #include <WiFiNl
27
28  void setup() {
```

WiFiNINA Scan Networks Advanced

This example scans for 802.11b/g network with one of the boards that support this library. Your Arduino Software (IDE) serial monitor will print out information about the board and the networks it can see, with the encryption type. It will not connect to a network.

Open your Arduino Software (IDE) serial monitor to view the networks the WiFi module can see. The shield may not see as many networks as your computer, but it will show also the encryption type.



```
1  /*
2
3   This example
4
5   scans for avai
6
7   Every ten seco
8
9   connect to any
10
11  BSSID and WiFi
12
13  Circuit:
14
15  * Board with N
16
17  This example i
18
19  created 1 Mar
20
21  by Arturo Guad
22
23  */
24
25  #include <SPI.h>
26  #include <WiFiN
27
28  void setup() {
```

WifiNINA Simple Web Server WiFi

In this example, a simple web server lets you blink an LED via the web. This example will print the IP address of your WiFi module (once connected) to the Arduino Software (IDE) serial monitor. Once you know the IP address of our board, you can open that address in a web browser to turn on and off the LED on pin 9.



If you do not have an LED, you can use the built-in LED instead (just remember to change out "9" to "LED_BUILTIN")

If the IP address of your shield is yourAddress:

<http://yourAddress/H>

turns the LED on

<http://yourAddress/L>

turns it off

This example is written for a network using WPA encryption. For WEP or WPA, change the Wifi.begin() call accordingly.

You should have access to a 802.11b/g wireless network that connects to the internet for this example. You will need to change the network settings in the sketch to correspond to your particular networks SSID.

For networks using WPA/WPA2 Personal encryption, you need the SSID and password. The shield will not connect to networks using WPA2 Enterprise encryption.

WEP network passwords are hexadecimal strings known as keys. A WEP network can have 4 different keys; each key is assigned a "Key Index" value. For WEP encrypted

```
1  /*
2
3    WiFi Web Serve
4
5    A simple web se
6
7    This sketch wil
8
9    to the Serial m
10
11   to turn on and
12
13   If the IP addre
14
15   http://yourAddr
16
17   http://yourAddr
18
19   This example is
20
21   WEP or WPA, cha
22
23   Circuit:
24
25   * Board with NI
26
27   * LED attached
28
```

WifiNINA Udp NTP Client

In this example, you will use your board's wifi capabilities to query a Network Time Protocol (NTP) server. In this way, your board can get the time from the Internet.

You should have access to a 802.11b/g wireless network that connects to the internet for this example. You will

correspond to your particular networks SSID.

```
1  /*
2
3  Udp NTP Client
4
5  Get the time fr
6
7  Demonstrates us
8
9  For more on NTP
10
11 see http://en.w
12
13 created 4 Sep 2
14
15 by Michael Marg
16
17 modified 9 Apr
18
19 by Tom Igoe
20
21 This code is in
22
23 */
24
25 #include <SPI.h>
26 #include <WiFiNl
27 #include <WiFiUc
28
```

WifiNINA WiFi Chat Server

A simple server that distributes any incoming messages to all connected clients. To use, open a terminal window, telnet to your WiFi module's IP address, and type away. Any incoming text will be sent to all connected clients

Additionally, you will be able to see the client's input in your Arduino Software (IDE) serial monitor as well.

You should have access to a 802.11b/g wireless network that connects to the internet for this example. You will need to change the network settings in the sketch to correspond to your particular networks SSID.

For networks using WPA/WPA2 Personal encryption, you need the SSID and password. The shield will not connect to networks using WPA2 Enterprise encryption.

WEP network passwords are hexadecimal strings known as keys. A WEP network can have 4 different keys; each key is assigned a "Key Index" value. For WEP encrypted networks, you need the SSID, the key, and key number.

```
1  /*
2
3  Chat  Server
4
5  A simple server
6
7  connected clien
8
9  You can see the
10
11 This example is
12
13 WEP or WPA, cha
14
15 Circuit:
16
17 * Board with NI
18
19 created 18 Dec
20
21 by David A. Mel
22
23 modified 31 May
24
25 by Tom Igoe
26
27 */
28
```

WifiNINA WiFi Ping

This example connects to a encrypted WiFi network (WPA/WPA2) then it prints the MAC address of the WiFi module, the IP address obtained, and other network details. After this initial phase, the loop continuously pings a given host, specified by IP Address or name.

You should have access to a 802.11b/g wireless network that connects to the internet

settings in the sketch to correspond to your particular networks SSID.

For networks using WPA/WPA2 Personal encryption, you need the SSID and password. The module will not connect to networks using WPA2 Enterprise encryption.

```
1  /*
2
3   This example c
4
5   Then it prints
6
7   the IP address
8
9   Then it contin
10
11  Circuit:
12
13  * Board with N
14
15  created 13 Jul
16
17  by dlf (Metodo
18
19  modified 09 Ju
20
21  by Petar Georg
22
23  */
24  #include <SPI.h>
25  #include <WiFiN
26
27  #include "arduin
28  //please en
```

WifiNINA WiFi SSL Client

transfers data using always SSL.

You should have access to a 802.11b/g wireless network that connects to the internet for this example. You will need to change the network settings in the sketch to correspond to your particular networks SSID.

For networks using WPA/WPA2 Personal encryption, you need the SSID and password. The module will not connect to networks using WPA2 Enterprise encryption.

```
1  /*
2
3  This example cre
4
5  data using alway
6
7  It is compatible
8
9  connections, lik
10
11 Written by Artur
12
13 last revision No
14
15 */
16
17 #include <SPI.h>
18 #include <Wi-Fi>
19
20 #include "arduin
21 //please en
22 char ssid[] = SE
23 char pass[] = SE
24 int keyIndex = 0
25
26 int status = WL_
27 // if you don't
28 // use the numer
```

WifiNINA WiFi Udp Send Receive String

This tutorial waits for a UDP packet on a local port. When a valid packet is received, an acknowledge packet is sent back to the client on a specified outgoing port. It relies on a WiFi connection made to your WiFi equipped board.

You should have access to a 802.11b/g wireless network that connects to the internet

settings in the sketch to correspond to your particular networks SSID.

```
1  /*
2
3    WiFi UDP Send
4
5    This sketch wai
6
7    When a packet i
8
9    created 30 Dece
10
11   by dlf (Metodo2
12
13   */
14
15   #include <SPI.h>
16   #include <WiFiN
17   #include <WiFiUa
18
19   int status = WL_
20   #include "arduin
21   //please en
22   char ssid[] = SE
23   char pass[] = SE
24   int keyIndex = 0
25
26   unsigned int loc
27
28   char packetBuffe
```

WifiNINA WiFi Web Client

This example shows you how to make a HTTP request using a WiFi module equipped board. It returns a [Google search for the term "Arduino"](#). The results of this search are viewable as HTML through your Arduino

This example is written for a network using WPA encryption. For WEP or WPA, change the `Wifi.begin()` call accordingly.

```
1  /*
2
3    Web client
4
5    This sketch con
6
7    using the WiFi
8
9    This example is
10
11   WEP or WPA, cha
12
13   This example is
14
15   WEP or WPA, cha
16
17   Circuit:
18
19   * Board with NI
20
21   created 13 July
22
23   by dlf (Metodo2
24
25   modified 31 May
26
27   by Tom Igoe
28
```

WifiNINA WiFi Web Client Repeating

This example shows you how to make repeated HTTP requests using a WiFi equipped board. It connects to <http://example.org>. The content of the page is

Arduino Software (IDE) Serial Monitor window.

You should have access to a 802.11b/g wireless network that connects to the internet for this example. You will need to change the network settings in the sketch to correspond to your particular networks SSID.

For networks using WPA/WPA2 Personal encryption, you need the SSID and password. The shield will not connect to networks using WPA2 Enterprise encryption.

WEP network passwords are hexadecimal strings known as keys. A WEP network can have 4 different keys; each key is assigned a "Key Index" value. For WEP encrypted networks, you need the SSID, the key, and key number.

```
1  /*
2
3    Repeating Wifi
4
5    This sketch con
6
7    using a WiFi eq
8
9    created 23 Apri
10
11   modified 31 May
12
13   by Tom Igoe
14
15   modified 13 Jan
16
17   by Federico Van
18
19   https://docs.ar
20
21   This code is in
22
23   */
24
25   #include <SPI.h>
26   #include <WiFiNINA
27
28   #include "arduin
```

WiFiNINA WiFi Web Server

In this example, you will use the WiFi module of your board to create a simple Web server. Using the WiFi 1010 library, your device will be able to answer a HTTP request received from the WiFi connection. After opening a browser and navigating to your WiFi's IP address, your board will respond with just enough HTML for a browser to

This example is written for a network using WPA encryption. For WEP or WPA, change the `Wifi.begin()` call accordingly.

You should have access to a 802.11b/g wireless network that connects to the internet for this example. You will need to change the network settings in the sketch to correspond to your particular networks SSID.

For networks using WPA/WPA2 Personal encryption, you need the SSID and password. The shield will not connect to networks using WPA2 Enterprise encryption.

WEP network passwords are hexadecimal strings known as keys. A WEP network can have 4 different keys; each key is assigned a "Key Index" value. For WEP encrypted networks, you need the SSID, the key, and key number.


```
1  /*
2    WiFi Web Serve
3
4    A simple web se
5
6    This example is
7    WEP or WPA, cha
8
9    Circuit:
10   * Analog inputs
11
12   created 13 July
13   by dlf (Metodo2
14   modified 31 May
15   by Tom Igoe
16
17   */
18
19   #include <SPI.h>
20   #include <WiFiNl
21
22
23   #include "arduin
24   //////////please en
25   char ssid[] = SE
26   char pass[] = SE
27   int keyIndex = 0
28
```

NINA Firmware

Firmware Version

It is important to check firmware version against WiFiNINA library version so that they are both aligned. Having different versions of library and firmware may create compatibility issues.

Code

When you load the sketch on the board, it will wait for a serial monitor console to be opened on your computer, then it prints out the result of the check between the expected firmware and the one available. If everything is fine, then you can proceed with your tasks, otherwise you should update the firmware following this update procedure.



Firmware version.

The sketch can be found in the snippet below:



```
1  /*
2
3  * This example c
4
5  * is updated.
6
7  *
8
9  * Circuit:
10
11  * - Board with N
12
13  *
14
15  * Created 17 Oct
16
17  * This code is i
18
19  */
20 #include <SPI.h>
21 #include <WiFiNINA
22
23 void setup() {
24
25     // Initialize s
26
27     Serial.begin(96
28
```

Firmware Update

To update the firmware on your NINA module, you can visit the

[Firmware/Certificate uploader for u-Blox NINA modules](#) tutorial for instructions.

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